Smart Health

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Project Overview: Smart Community Health

1. Problem Statement

Rural and tribal communities in India, particularly in the North East region (Assam/North East focus), face significant healthcare challenges:

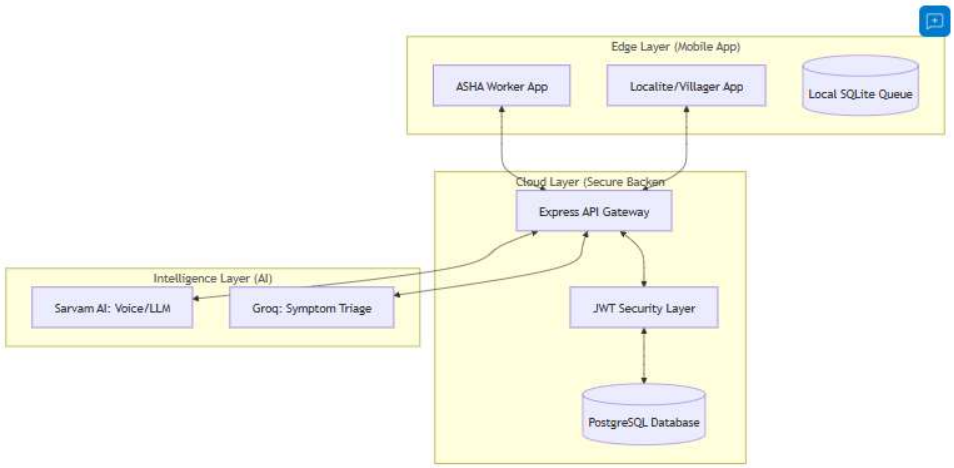
- **Data Scarcity**: Lack of real-time health data for proactive disease surveillance.
- **Infrastructure Gaps**: Limited access to formal healthcare facilities and specialists.
- **Environmental Risks**: Unmonitored water safety leading to water-borne diseases.
- **Communication Barriers**: Inefficient paper-based reporting by frontline health workers (ASHA).
- **Awareness Deficit**: Villagers often lack immediate, localized health advice and precautions.

2. Our Approach

We have developed **Smart Community Health**, a comprehensive, AI-driven digital health ecosystem. The system follows a **Hub-and-Spoke model**:

- **Edge (Mobile App)**: Primary data collection by ASHA workers and Localites.
- **Cloud (Backend Server)**: Centralized orchestration of AI, analytics, and data storage.
- **Intelligence Layer**: Integration of cutting-edge AI (Sarvam and Groq) for triage, transcription, and health insights.

2.1. System Architecture



Design Philosophy:

- **Offline-First**: Reliable data entry and access in zero-connectivity zones using local SQLite queues.
- **Multilingual & Voice-First**: Inclusive interface for diverse linguistic backgrounds using Voice AI.
- **Proactive Monitoring**: Surveillance engine to detect disease clusters before they become outbreaks.

3. Tech Stack

- **Frontend**: React Native (Expo) - Cross-platform mobile development for iOS and Android.
- **Backend**: Node.js (Express) - Robust and scalable API gateway.
- **Database**:
  - **PostgreSQL**: Cloud-based source of truth for all health data.
  - **SQLite**: On-device caching for offline reliability.
- **AI Engines**:
  - **Sarvam AI**: Multilingual Voice-to-Text (STT), Text-to-Speech (TTS), and LLM for the "Simran" Voice Assistant.
  - **Groq (Llama-3)**: High-speed analysis for symptom triage and cluster detection.
- **Storage**: Firebase Cloud Storage for images and audio files.
- **UI/UX**: React Native Paper for a professional "Midnight Blue & Gold" premium aesthetic.

4. Key Features

For ASHA Workers (Health Warriors)

- **Digital Kit**: Paperless symptom and water quality reporting with geo-tagging.
- **Cluster Detection**: Real-time alerts for emerging infection zones (e.g., >5 fever cases in 72 hours).
- **Offline Queue**: Seamless operation without internet; automatic syncing when online.

For Localites (The Community)

- **Simran Voice Assistant**: A multilingual AI assistant for immediate health guidance and assistance requests.
- **Water Safety Dashboard**: Real-time safety status of village water sources (e.g., pH, turbidity).

- **Community Alerts**: AI-generated precautions based on local data (e.g., "Boil water alert").

For Clinic Staff (Decision Makers)

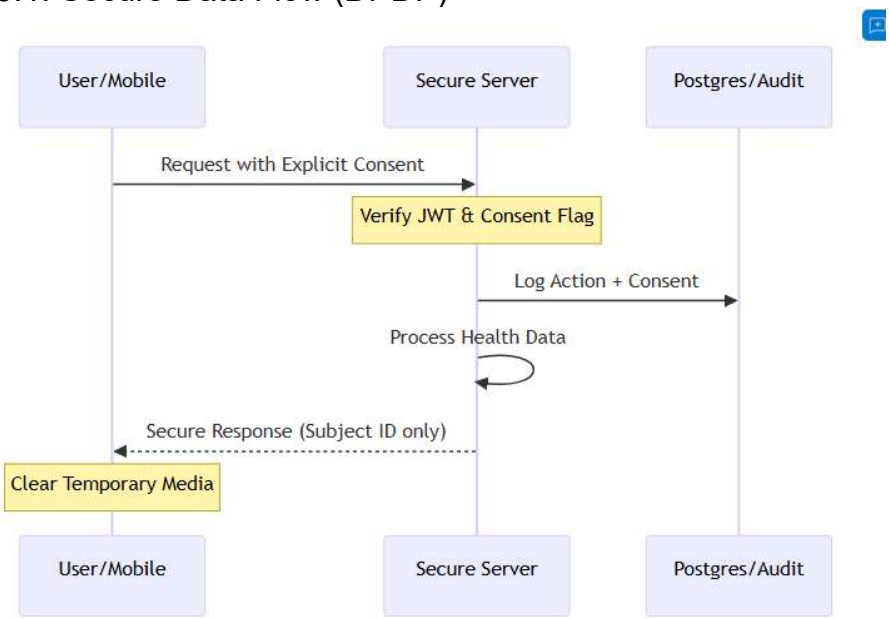
- **Outbreak Dashboard**: High-level view of village health trends and active clusters.
- **Resource Allocation**: Data-driven assignment of follow-ups and medical supplies.

5. Data Privacy & Security (DPDP Compliance)

The platform is designed to be **DPDP-ready** (Digital Personal Data Protection Act, India), featuring multiple layers of security:

- **Explicit Consent Management**: Mandatory digital consent flows for collecting and processing symptoms, geo-tags, and audio data.
- **Data Minimization**: Using hardened JWT authentication where tokens contain only essential subject identifiers, keeping sensitive health data strictly on the server or in secure storage.
- **Audit Trail**: Comprehensive logging of all sensitive operations (who accessed what data and when) to ensure accountability.
- **Data Retention & Erasure**: Automated anonymization of old records and built-in "Right to Erasure" flows for user data deletion.
- **On-Device Privacy**: Hardware-backed secure storage for keys and automatic purging of temporary media files post-upload.

5.1. Secure Data Flow (DPDP)



6. Why Choose This Project?

1. **Scalable Impact**: Directly addresses the health needs of underserved communities.
2. **Innovative AI Integration**: Uses state-of-the-art voice and logic AI to bridge the digital divide.
3. **Resilient Design**: Built to work in the toughest environments (offline-first).
4. **End-to-End Solution**: From environmental monitoring (water) to clinical follow-up.
5. **DPDP Compliance**: Built with India's latest data protection laws in mind, ensuring trust and legality in sensitive health environments.